

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

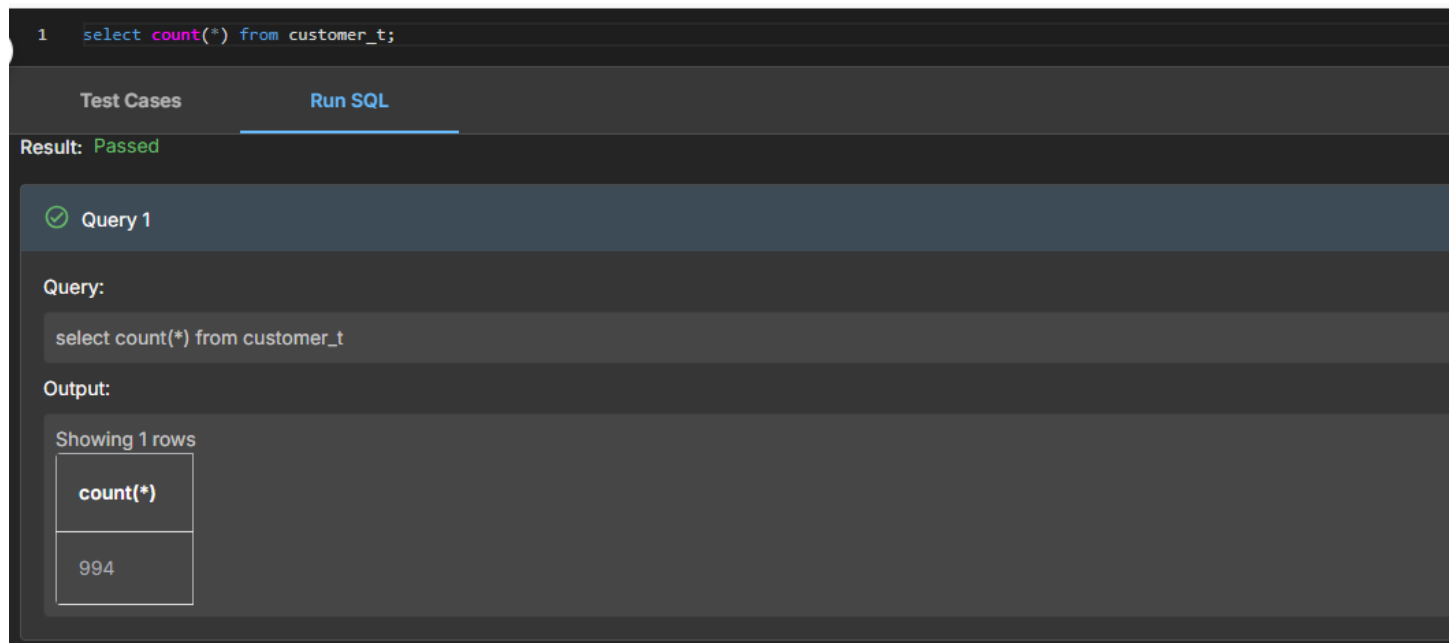
As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

1.1 Solution Query:

```
select count(*) from customer_t;
```

Output:



The screenshot shows a SQL query execution interface. At the top, there is a text input field containing the query: `1 select count(*) from customer_t;`. Below this, there are two tabs: "Test Cases" and "Run SQL". The "Run SQL" tab is selected. Below the tabs, the result is displayed as "Result: Passed". Underneath, there is a section titled "Query 1" with a green checkmark icon. Below this, the query is repeated: `select count(*) from customer_t`. Below the query, the output is shown: "Showing 1 rows". A table with one column "count(*)" and one row containing the value "994" is displayed.

count(*)
994

1.2: Solution Query:

```
select state, count(*) from customer_t
```

```
group by 1
```

```
order by 2;
```

Output:

Test Cases
Run SQL

Result: **Passed**


Query 1

Query:

```
select state, count(*) from customer_t
group by 1
order by 2
```

state	count(*)
Maine	1
Vermont	1
Wyoming	1
Mississippi	2
North Dakota	2
Montana	3

Observations and Insights:

- The total number of customers who has placed order in New-Wheels is 994
- The distribution of customers varies from state to state; the above output shows the distribution of first few states.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT p.vehicle_maker, count(o.product_id) as order_count FROM
customer_t c

join order_t o on c.customer_id = o.customer_id

join product_t p on o.product_id = p.product_id

group by p.vehicle_maker

order by 2 desc

limit 5;
```

Output:

```
1 SELECT p.vehicle_maker, count(o.product_id) as order_count FROM
2 customer_t c
3 join order_t o on c.customer_id = o.customer_id
4 join product_t p on o.product_id = p.product_id
5 group by p.vehicle_maker
6 order by 2 desc
7 limit 5
```

Test Cases Run SQL

Result: Passed

✓ Query 1

Output:

Showing 5 rows

vehicle_maker	order_count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- From the above output, it is well known that many customers prefer the maker Chevrolet followed by Ford, Toyota and so on.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```

1  with top_ranked_makers as (
2      SELECT c.state, p.vehicle_maker, count(o.product_id) as order_count,
3          rank() over (partition by c.state order by count(o.product_id) DESC) as ranking
4  FROM customer_t c
5  join order_t o on c.customer_id = o.customer_id
6  join product_t p on o.product_id = p.product_id
7  group by c.state, p.vehicle_maker
8  )
9  SELECT state, vehicle_maker, order_count, ranking
10 FROM top_ranked_makers
11 WHERE ranking = 1
12 ORDER BY state;

```

Test Cases

Run SQL

Result: Passed

✓ Query 1

Output:

	state	vehide_maker	order_count	ranking
▶	Alabama	Dodge	5	1
	Alaska	Chevrolet	2	1
	Arizona	Pontiac	3	1
	Arizona	Cadillac	3	1
	Arkansas	Suzuki	1	1
	Arkansas	Chevrolet	1	1
	Arkansas	Pontiac	1	1
	Arkansas	Volkswaen	1	1

Observations and Insights:

- It is observed that the preferred vehicle maker varies from state to state.
- It is observed that the top five vehicle makers compete with each other in each state.

Question 4: Find the overall average rating given by the customers.
What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
1  SELECT quarter_number, Round(avg(case
2      when customer_feedback = 'Very Bad' then 1
3      when customer_feedback = 'Bad' then 2
4      when customer_feedback = 'Okay' then 3
5      when customer_feedback = 'Good' then 4
6      when customer_feedback = 'Very Good' then 5
7      else null
8      end),2) as feedback_ratings
9  from order_t
10 group by quarter_number;
```

Test Cases

Run SQL

Result: Passed

✓ Query 1

Output:

Output:

Showing 4 rows

quarter_number	feedback_ratings
1	3.55
2	3.35
3	2.96
4	2.4

Observations and Insights:

- According to the feedback given by customers the ratings begin well during the first quarter then it has lowered significantly at the fourth quarter.
- The company has to focus on what changes has led the rating dip so low from 1st quarter to the 4th quarter.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
1 select quarter_number,
2 Round(
3 (Count(case when customer_feedback = 'Very Bad' then 1 END) * 100.0)/ count(*), 2) as very_bad_precentage,
4 Round(
5 (Count(case when customer_feedback = 'Bad' then 2 END) * 100.0)/ count(*), 2) as bad_precentage,
6 Round(
7 (Count(case when customer_feedback = 'Okay' then 3 END) * 100.0)/ count(*), 2) as okay_precentage,
8 Round(
9 (Count(case when customer_feedback = 'Good' then 4 END) * 100.0)/ count(*), 2) as good_precentage,
10 Round(
11 (Count(case when customer_feedback = 'Very Good' then 5 END) * 100.0)/ count(*), 2) as very_good_precentage
12 from order_t
13 group by quarter_number
14 order by quarter_number;
```

Test Cases

Run SQL

Result: Passed

✓ Query 1

Output:

Output:

Showing 4 rows

quarter_number	very_bad_precentage	bad_precentage	okay_precentage	good_precentage	very_good_precentage
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- It is observed that the first quarter had about forty percentage of good reviews, whereas in the 4th quarter there is only about 20 percentage of good reviews, which is very low.
- New-Wheels has to focus on why the customer's feedback has changed over the time and what services could make the customers feel satisfied to get better reviews.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
1 select quarter_number, count(order_id)
2 from order_t
3 group by quarter_number
4 order by quarter_number;
```

Test Cases

Run SQL

Result: Passed

✓ Query 1

Output:

Output:

Showing 4 rows

quarter_number	order_count
1	310
2	262
3	229
4	199

Observations and Insights:

- The number of orders for the first quarter was 310 and for the fourth quarter is 199.
- This shows that number of orders is significantly low
- The company has to address to the reasons as to why the number of orders has reduced.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
1 SELECT SUM(quantity * (vehicle_price - discount)) AS net_revenue
2 FROM order_t;
3
```

Test Cases

Run SQL

Result: Passed

✓ Query 1

```
1 SELECT
2     quarter_number,
3     SUM(quantity * (vehicle_price - discount)) AS current_quarter_revenue,
4     LAG(SUM(quantity * (vehicle_price - discount)), 1) OVER (ORDER BY quarter_number) AS previous_quarter_revenue,
5     CASE
6         WHEN LAG(SUM(quantity * (vehicle_price - discount)), 1) OVER (ORDER BY quarter_number) IS NOT NULL
7         THEN
8             ((SUM(quantity * (vehicle_price - discount)) -
9              LAG(SUM(quantity * (vehicle_price - discount)), 1) OVER (ORDER BY quarter_number)) /
10             LAG(SUM(quantity * (vehicle_price - discount)), 1) OVER (ORDER BY quarter_number)) * 100
11         ELSE NULL
12     END AS quarter_over_quarter_percentage_change
13 FROM order_t
14 GROUP BY quarter_number
15 ORDER BY quarter_number;
```

Test Cases

Run SQL

Result: Passed

✓ Query 1

Output:

Output:

Showing 1 rows

net_revenue

125481878.22999999

Output:

Showing 4 rows

quarter_number	current_quarter_revenue	previous_quarter_reve...	quarter_over_quarter_...
1	39637378.160000026		
2	32913497.439999998	39637378.160000026	-16.963485054078163
3	29435188.489999995	32913497.439999998	-10.568032024979434
4	23495814.140000004	29435188.489999995	-20.17780301294069

Observations and Insights:

- The net revenue of the company generated is 125481878.2299
- The change in revenue from the subsequent quarter to the previous quarter is as seen above. This sums that the revenue decreases in each quarter.
- This shows that the company's sales and profit has been lowered across each subsequent quarter.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
1 select quarter_number, sum(vehicle_price) as net_revenue, count(order_id) as Quarter_wise_count
2 from order_t
3 group by 1
4 order by 1;
```

Test Cases

Run SQL

Result: Passed

✓ Query 1

Output:

Output:

Showing 4 rows

quarter_number	net_revenue	Quarter_wise_count
1	26519199.190000001	310
2	21595874.350000013	262
3	19719917.590000004	229
4	15280009.979999995	199

Observations and Insights:

- The trend of net revenue and quarter wise count decreases for subsequent quarter.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
1  select c.credit_card_type, avg(o.discount) as Average_disc_offered
2  from customer_t c
3  join order_t o
4  on c.customer_id = o.customer_id
5  group by c.credit_card_type;
```

Test Cases

Run SQL

Result: Passed

✓ Query 1

Output:

Showing first 10 rows out of 16 rows

credit_card_type	Average_disc_offered
americanexpress	0.616326530612245
bankcard	0.6095454545454548
china-unionpay	0.6221739130434784
diners-club-carte-blanc	0.6144897959183674
diners-club-enroute	0.5997916666666666
diners-club-international	0.584

Observations and Insights:

- The average discount offered for different types of credit cards ranges from 0.6 to 0.5

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
1 select round(avg(julianday(ship_date) - julianday(order_date)), 2) as Avg_time_taken, quarter_number
2 from order_t
3 group by quarter_number
4 order by quarter_number;
```

Test Cases

Run SQL

Result: Passed

✓ Query 1

Output:

Output:

Showing 4 rows

Avg_time_taken	quarter_number
57.17	1
71.11	2
117.76	3
174.1	4

Observations and Insights:

- The average time take to ship the order in the first quarter is 57.17 and increases over each quarter. The avg time taken to ship the order in the 4th quarter is 174.1
- There is a significant difference in the avg time taken to ship from the 1st quarter to the 4th quarter
- This could impact the business in various levels.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
83115001.11	1000	994	3.13
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
15280009.98	199	97.96	21.50

Business Recommendations

- The quarterly report with all the key metrics shows the dip in sales over the time.
- The company should focus more on the top five vehicle makers preferred by the customers to improve sales.
- The company should minimize the time taken to ship the order, which could improve sales and feedback from customers.
- Promoting the company in areas where the distribution of customers is relatively low could help in attracting new customers.
- An analysis should be conducted on the segment of customers who gave low feedback to identify common factors and determine which points contributed to the low feedback.
- An analysis should be conducted on why sales and customer feedback changed over the time.